Information Rates Over DMCs With Many Independent Views

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***Abstract*—In this paper, we investigate the fundamental limits of reliable communication over a discrete memoryless channel (DMC) when there are a large number of noisy views of a transmitted symbol, i.e., when several copies of a single symbol are sent independently through the DMC. We argue that the channel capacity and dispersion of such a *multi-view* DMC converge exponentially quickly in the number of views to to the entropy and varentropy of the input distribution, respectively, and identify the exact rate of convergence. This rate equals the smallest Chernoff information between two conditional distributions of the output given unequal inputs. We also present a new channel model that we call the *Poisson approximation channel*—of possible independent interest—whose capacity closely approximates the capacity of the multi-view binary symmetric channel (BSC).**

I. Introduction

Consider a communication setting where the decoder receives multiple, potentially noisy, views of a single transmitted sequence that is sent via independent transmissions through a noisy channel. Such a scenario arises naturally, for example, in DNA-based storage systems (see, e.g., [1]–[4]). In such systems, the (short) sequenced DNA molecules are "amplified" by Polymerase Chain Reaction (PCR), to produce several copies of each molecule. The resultant pool of molecules is then sampled from, and the sampled molecules subjected to noisy reads, during sequencing. Clearly, multiple noisy views (or multiple draws) of a transmitted sequence can only improve the error probabilities of decoding, as compared to decoding a single received sequence. Viewed differently, it is clear that the Shannon capacity of the *multi-view* channel, as described above, is at least as large as the capacity of the standard single-view channel.

In this paper, we restrict our attention to the setting where the noisy channel is a discrete memoryless channel (DMC). The capacities of multi-view DMCs were studied first in [5], wherein the capacity of the multi-view binary symmetric channel (BSC) was identified and an explicit expression was provided. More generally, a study of the information rates over multi-view DMCs was carried out in the literature on "information combining" (see [6], [7] and the references therein), where bounds were obtained on the capacities of such channels. We also mention the closely related classic work of Levenshtein on sequence reconstruction [8], which dealt with the number of *distinct* noisy views required to guarantee exact recovery of the transmitted sequence, assuming combinatorial error models. Furthermore, for a DNA-based storage model with combinatorial errors, constructions of codes and bounds on their sizes can be found in works such as [9], [10]. Also related is the well-studied problem on trace reconstruction [11]–[13], wherein an input string is passed independently through a deletion channel and the goal is to find the number of resultant outputs (or "traces") required for high-probability reconstruction.

In this work, we consider the setting where the number of views of the transmitted sequence is very large, and study the asymptotic rate of growth of the information rate and the channel dispersion of general multi-view DMCs, under arbitrary input distributions. The generalization to arbitrary (non-binary input, asymmetric) DMCs assumes significance in the context of DNA-based storage, where the inputs are drawn from a quarternary alphabet and the DMC that models sequencing errors is asymmetric [14]. In particular, we show that the information rate (and hence the capacity), as well as the channel dispersion, converge exponentially quickly (in the number of views) to the input entropy and varentropy, respectively, and identify the exact rate of convergence. For DNA-based storage channels, for example, following [4, Thm. 5], we see that understanding the rate of convergence of the information rate to its limit helps provide an upper bound on the number of times the stored DNA molecules must be sampled, as a function of their length, so that the resulting channel (arising due to sampling and sequencing) is almost noiseless.

Further, for the special case of the multi-view BSC, we explore a closely related channel, which we call the *Poisson approximation channel*, of possible independent interest, whose capacity is a tight lower bound on the capacity of the multi-view BSC.

The paper is organized as follows: Section II sets down the notation; Section III presents our main results; Section IV provides a proof of the rate of convergence of the information rate to the input entropy; Section V provides a proof of the rate of convergence of the channel dispersion to the input varentropy, and Section VI contains a proof that the capacity of the Poisson approximation channel is a tight lower bound on the capacity of the multi-view BSC. Section VII concludes the paper and proposes a few problems for further research.

II. Notation

All logarithms are assumed to be natural logarithms. Random variables are denoted by capital letters, e.g., X, Y, and small letters, e.g., x, y, denote their instantiations. Sets are denoted by calligraphic letters, e.g., $\mathcal{X}, \mathcal{Y}$. The set of positive natural numbers $\{1, 2, \ldots\}$ is denoted as $\mathbb{N}$. Notation

such as $P(x), P(y|x)$ are used to denote the probabilities $P_X(x), P_{Y|X}(y|x)$, when it is clear which random variables are being referred to. The notation $\mathsf{H}(X) := \mathbb{E}[-\log P(X)]$ and $\mathsf{V}(X) := \mathbb{E}\left[(-\log P(X) - \mathsf{H}(X))^2\right]$ denote the entropy and varentropy of X, which depend only on the distribution of X. The Chernoff information $\mathsf{C}(P, Q)$ between two distributions P and Q defined on the same alphabet $\mathcal{X}$ is given by (see [15, Ch. 11])

$$\mathsf{C}(P, Q) := -\min_{\lambda \in [0,1]} \log\left(\sum_{x \in \mathcal{X}} P(x)^{1-\lambda} Q(x)^{\lambda}\right). \tag{1}$$

Further, the standard Kullback-Leibler (KL) divergence between two distributions P, Q on the same alphabet $\mathcal{X}$ is given by $D(P||Q) := \mathbb{E}_{X \sim P}[\log \frac{P(X)}{Q(X)}]$, where we take $0 \log \frac{0}{z} = 0$ and we set $D(P||Q) = \infty$ if there exists $x \in \mathcal{X}$ such that $Q(x) = 0$ but $P(x) > 0$. The notation a^n is used to denote the length-n vector $(a_1, \ldots, a_n)$.

For a symbol b, the notation $\mathbb{1}\{x = b\}$ is used to denote the indicator that x equals b; this indicator equals 1 if $x = b$ and 0, otherwise. For sequences $(a_n)_{n \geq 1}$ and $(b_n)_{n \geq 1}$ of positive reals, $a_n = o(b_n)$ if $\lim_{n \to \infty} \frac{a_n}{b_n} = 0$, that $a_n = O(b_n)$ if for all n large enough, $a_n \leq c \cdot b_n$, for some $c > 0$, and that $a_n = \Theta(b_n)$ if for all n large enough, we have that $m \cdot b_n \leq a_n \leq M \cdot b_n$, for some constants $m, M \in \mathbb{R}$.

III. Main Results

We now formalize the problem and present our main results. Consider a DMC W with finite input and output alphabets $\mathcal{X}$ and $\mathcal{Y}$, respectively, specified by the channel law $\{P_{Y|X}(y|x) = W(y|x) : \ x \in \mathcal{X}, y \in \mathcal{Y}\}$. Let $X \sim P_X$ be the input to the channel. Our setting of interest is when $d \geq 1$ noisy outputs are obtained by independently passing the input through the DMC. Let these outputs be denoted by $Y_1, \ldots, Y_d$; their conditional distribution obeys

$$P(Y^d = y^d \mid X = x) = \prod_{i=1}^{d} W(y_i|x_i).$$

We call a DMC with a channel law as above as the *d-view* DMC $W^{(d)}$. We assume further that $|\mathcal{X}|$ and $|\mathcal{Y}|$ do not depend on d.

Now, given any DMC W, we denote its capacity by $C(W) = \max_{P_X} I(X;Y) = \max_{P_X} \mathbb{E}\left[\log \frac{W(Y|X)}{\sum_{x \in \mathcal{X}} P(x) W(Y|x)}\right]$ [16]. Hence, the capacity of $W^{(d)}$ is given by $C^{(d)} := \max_{P_X} I^{(d)}$, where $I^{(d)} := I(X; Y^d)$. Further, for any input distribution P_X, the channel dispersion of the d-view DMC W is given by

$$V^{(d)} := \mathbb{E}\left[\left(\iota(X; Y^d) - I(X; Y^d)\right)^2\right],$$

where $\iota(x; y^d) := \log \frac{P(y^d|x)}{P(y)}$. We are interested in characterizing the behaviour of $C^{(d)}$ (or more generally, of $I^{(d)}$) and $V^{(d)}$ when d is large. From operational considerations, we can argue that $I^{(d)}$ and $V^{(d)}$ are non-decreasing in d (more formally, for $I^{(d)}$, the data processing inequality [15, Thm. 2.8.1] guarantees that $I^{(d)}$ is non-decreasing in d). Furthermore, it is reasonable to expect that $I^{(d)}$ and $V^{(d)}$ converge to $\mathsf{H}(X)$ and $\mathsf{V}(X)$, respectively, since for very large d, the "uncertainty" introduced by the channel is expected to drop to 0. In this work, we confirm these convergence results by showing that the $I^{(d)}$ and $V^{(d)}$ in fact converge to their limiting values exponentially quickly in d, and explicitly identify the rate of convergence.

To build some intuition why the convergence is exponentially fast in d, at least for $I^{(d)}$, consider the simple example of the d-view BSC(p), which we denote by BSC$^{(d)}(p)$ (here, $p < 1/2$ is the crossover probability of the BSC), with a uniform input distribution. At the decoder end, we construct a simple "data processor" that takes the (random) outputs $Y^d \in \{0,1\}^d$ and computes their majority $M \in \{0,1\}$. Then,

$$\begin{aligned} I^{(d)} &= I(X; Y^d) \\ &\geq I(X; M) = 1 - h_b(\Pr[X \neq M]), \end{aligned}$$

where the inequality holds by the data processing inequality and the last equality holds by symmetry. Now, observe that since $p < 1/2$, we have by the Chernoff bound (see [17, Example 1.6.2]) that

$$\begin{aligned} \Pr[X \neq M] &\leq \exp(-d \cdot D(\mathrm{Ber}(1/2)||\mathrm{Ber}(p))) \\ &= \exp(-d \cdot Z(p)), \end{aligned}$$

where $Z(p) = 2\sqrt{p(1-p)}$ is the Bhattacharya parameter for the BSC(p). Hence, since $h_b(\gamma) \leq -\gamma \log \gamma$, for all $\gamma \in [0,1]$, we see that $I^{(d)} \geq 1 - dZ(p) \cdot \exp(-d \cdot Z(p))$, thereby providing reason for the intuition that the convergence of $I^{(d)}$ to 1 is exponentially fast in d (in fact, in Theorem III.1, we argue that the rate $Z(p)$ is in fact tight for the BSC(p)).

Our main theorem is given below.

Theorem III.1. *We have that*

$$\begin{aligned} I^d &= H(X) - e^{-d \cdot \rho + \Theta(\log d)}, \text{ and} \\ V^d &= e^{-d \cdot \rho + \Theta(\log d)}, \end{aligned}$$

where $\rho = \min_{x, x' : x \neq x'} \mathsf{C}(P_{Y|x}, P_{Y|x'})$.

We proceed to interpret this result. Suppose we define the rates

$$\begin{aligned} \rho(I) &:= \liminf_d -\frac{1}{d} \cdot \log\left(\mathsf{H}(X) - I^{(d)}\right) \\ &= \liminf_d -\frac{1}{d} \cdot \log \mathsf{H}(X|Y^d), \text{ and} \\ \rho(V) &:= \liminf_d -\frac{1}{d} \cdot \log\left(\mathsf{V}(X) - V^{(d)}\right), \end{aligned}$$

we get by the above result that these rates of convergence obey $\rho(I) = \rho(V) = \rho$, which in turn is independent of the input distribution P_X, with possible dependence on its support. Further, the limits in the definitions of these rates exist. Further, for a binary-input memoryless symmetric (BIMS) channel W (see [18, Chap. 4] for a definition), we have that the function $f(\lambda) := \log\left(\sum_{y \in \mathcal{Y}} P(y|0)^{1-\lambda} P(y|1)^{\lambda}\right)$ is symmetric about $\lambda = 1/2$. Hence, using the fact that f is convex, we obtain that the minimum in (1) is attained at $\lambda = 1/2$, giving

$$\begin{aligned} \mathsf{C}(P_{Y|0}, P_{Y|1}) &= -\log \sum_{z \in \mathcal{Z}} \sqrt{P_{Y|0}(z|0) P_{Y|1}(z|1)} \\ &= -\log Z_b(W), \end{aligned}$$

where $Z_b(W)$ is the Bhattacharya parameter of the BIMS channel W. Hence, following the results in [19], the rates of convergence for a BIMS channel W are extremized by a BEC and BSC having the same capacity as W, respectively, with the BEC having the largest and BSC having the smallest Bhattacharya parameters, respectively. Here, for the d-view BSC(p), we have $\rho(I) = \rho(V) = -\log 2\sqrt{p(1-p)}$, and for the d-view BEC(ϵ), we have $\rho(I) = \rho(V) = -\log \epsilon$. Furthermore, for the Z-channel with parameter δ, $\rho(I) = \rho(V) = -\log \delta$. In fact, for most well-behaved DMCs W, we will have that $\rho(I) = \rho(V) > 0$, implying exponential convergence of mutual information and dispersion to $\mathsf{H}(X)$ and $\mathsf{V}(X)$, respectively.

Theorem III.1 along with the normal approximation for finite blocklengths [20] then allow us to characterize, up to exponential tightness, the largest finite-blocklength rates achievable over a d-view DMC $W^{(d)}$, under some regularity conditions. Our presentation follows that in [21] (see also [22, Thm. 22.2]). Suppose that the DMC $W^{(d)}$ is non-singular (see the definition in [21, Sec. 4.2.1]). Let $M^\star(n,\epsilon)$ be the largest integer M such that there exists a blocklength-n code with maximal probability of error $\epsilon > 0$ over $W^{(d)}$. Further, let $P_X^\star$ be that capacity-achieving input distribution for $W^{(d)}$ that minimizes $V^{(d)}$; we assume that $P_X^\star$ is independent of d and that under this distribution, $V^{(d)} > 0$. For $t \in \mathbb{R}$, let $Q(t) = \int_t^\infty \frac{1}{\sqrt{2\pi}} e^{-z^2/2} \mathrm{d}z$, with Q^{-1} denoting its inverse. We then have the following theorem:

Theorem III.2. *We have that for any non-singular DMC $W^{(d)}$, for all $n \geq 1$ and $\epsilon \in (0,1)$, under the capacity-achieving distribution $P_X^\star$,*

$$\frac{\log M^\star(n,\epsilon)}{n} = H(X) - e^{-d\rho+\Theta(\log d)} + Q^{-1}(\epsilon)\cdot\frac{e^{-d\rho/2+\Theta(\log d)}}{\sqrt{n}} + O\left(\frac{\log n}{n}\right),$$

where $\rho = \rho(I) = \rho(V)$.

Clearly, it follows from the above theorem that for any finite n, by choosing $d = \rho^{-1}\log n$, we can obtain rates up to $\frac{\log M^\star(n,\epsilon)}{n} = H(X) - O\left(\frac{\log n}{n}\right)$ over any d-view DMC $W^{(d)}$.

Our next contribution concerns the special case of the d-view BSC(p), which we denote by BSC$^{(d)}(p)$; the capacity of this channel is known to be equal to the capacity of the so-called binomial channel [5, Sec. III.A], denoted by $\mathsf{Bin}_d(p)$. This capacity is given by

$$C(\mathsf{Bin}_d(p)) = 1 + \sum_{i=0}^{d}\binom{d}{i}\left(p^i(1-p)^{d-i}\log\frac{p^i(1-p)^{d-i}}{p^i(1-p)^{d-i}+p^{d-i}(1-p)^i}\right), \tag{2}$$

which is achieved when the input distribution $P_X = \mathrm{Ber}(1/2)$. While Theorem III.1 determines the rate of convergence of $C\left(\mathrm{BSC}^{(d)}(p)\right) = C(\mathsf{Bin}_d(p))$ (via the rate of convergence of the mutual information $I(X;Y^d)$ under a uniform input distribution $P_X = \mathrm{Ber}(1/2)$) to $H(P_X) = 1$, it will be useful if a more fine-grained understanding of the capacity is obtainable. To this end, we identify a closely-related DMC, which we call the *Poisson approximation channel* and denote by $\mathsf{Poi}_d(p)$, whose capacity turns out to be a good approximation to $C(\mathsf{Bin}_d(p))$. The DMC $\mathsf{Poi}_d(p)$ has input alphabet $\mathcal{X} = \{0,1\}$ and output alphabet $\mathcal{R} = (\mathbb{N}\cup\{0\})\times(\mathbb{N}\cup\{0\})$. We denote the input bit as X and the output as the pair (R_1, R_2). The channel law obeys

$$P_{R_1,R_2|X}(r_1,r_2|x) = P_{R_1|X}(r_1|x)P_{R_2|x}(r_2|x),$$
$$\text{for all } x \in \{0,1\} \text{ and } (r_1,r_2) \in (\mathbb{N}\cup\{0\})\times(\mathbb{N}\cup\{0\}),$$

where

$$P_{R_1|0} = \mathrm{Poi}(d(1-p)),\ P_{R_2|0} = \mathrm{Poi}(dp),\ \text{and}$$
$$P_{R_1|1} = \mathrm{Poi}(dp),\ P_{R_2|1} = \mathrm{Poi}(d(1-p)).$$

Here, $\mathrm{Poi}(\gamma)$ denotes the Poisson distribution with parameter γ. Note that $\mathsf{Poi}_d(p)$ is a BIMS channel; hence, its capacity is achieved by $P_X = \mathrm{Ber}(1/2)$ (see [18, Problem 4.8]). The following theorem then holds.

Theorem III.3. *We have that*

$$C(\mathsf{Poi}_d(p)) \leq C(\mathsf{Bin}_d(p)) \leq C(\mathsf{Poi}_d(p)) + \exp(-d(1-Z(p))) - Z(p)^{2d},$$

where $Z(p) = 2\sqrt{p(1-p)}$ is the Bhattacharya parameter of the BSC(p).

Figure 1 plots the capacities of the $\mathsf{Bin}(p)$ and $\mathsf{Poi}(p)$ channels when $d = 24$. It is clear from the plots and from Theorem III.3 that $C(\mathsf{Poi}_d(p))$ is a tight lower bound on $C(\mathsf{Bin}_d(p))$.

We now discuss a simple extension of Theorem III.3 to d-view BIMS channels $W^{(d)}$ with a finite output alphabet. It is well-known (see, e.g., Section II in [23]) that any BIMS channel W with a finite output alphabet can be decomposed into finitely many BSC subchannels, i.e.,

$$W_{Y|X=x} = \sum_{i=1}^{K}\epsilon_i \cdot W_{Y|X=x}^{(i)},$$

for some $K < \infty$, for any $x \in \{0,1\}$, with $W_{Y|X=x}^{(i)}$ being the channel law of a BSC with crossover probability p_i. Here, ϵ_i is the probability of choosing subchannel i, with $\sum_i \epsilon_i = 1$. Let P_s be the distribution on $\{1,\ldots,K\}$ with mass ϵ_i at point i. We then have that

$$C^{(d)}(W) = \mathbb{E}\left[C(J_1,\ldots,J_d)\right], \tag{3}$$

where the random variables J_i, $1 \leq i \leq K$, are drawn i.i.d. according to P_s, and the notation $C(j_1,\ldots,j_d)$ refers to the capacity of that channel with the channel law

$$P(Y^d = y^d|X = x) = \prod_{\ell=1}^{d} W^{(j_\ell)}(y_\ell|x),$$

for $x \in \{0,1\}$ and $y^d \in \mathcal{Y}^d$. Now, suppose that $0 < p_1 \leq p_2 \leq \ldots p_K < 1/2$. We then have from Theorem III.3 and (3) that

$$C(\mathsf{Poi}_d(p_K)) \leq C^{(d)}(W) \leq C(\mathsf{Poi}_d(p_1)) + \exp(-d(1-Z(p_1))) - Z(p_1)^{2d}.$$

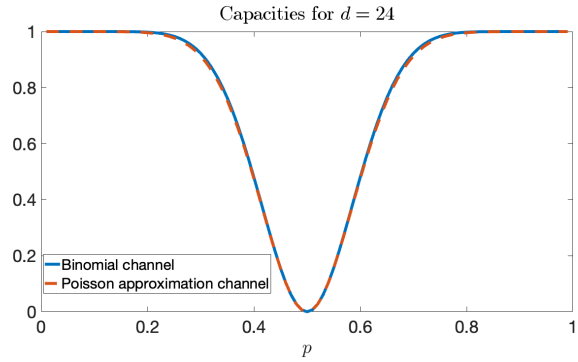

Fig. 1: Plots comparing $C(\mathsf{Bin}(p))$ and $C(\mathsf{Poi}(p))$ for $p \in [0, 1]$, when $d = 24$.

IV. Convergence of Mutual Information

In this section, we prove Theorem III.1 for the mutual information (or information rate) $I^{(d)}$ of a DMC W with an aribitrary distribution P_X.

A. Convergence of Mutual Information for BIMS Channels

To build intuition, we present a simple proof for the case when the DMC W is a BIMS channel, with $P_X = \mathrm{Ber}(1/2)$. The choice of the input distribution being uniform assumes relevance since the capacity of a BIMS channel is achieved by uniform inputs (see [18, Problem 4.8]).

The following bounds on the conditional entropy of the input to a BIMS channel given the output, which are drawn from [24, Prop. 2.8], will prove useful. Recall the definition of the Bhattacharya parameter of a BIMS channel W with output alphabet $\mathcal{Y}$ given by $Z_b(W) := \sum_{y \in \mathcal{Y}} \sqrt{P_{Y|0}(y) P_{Y|1}(y)}$.

Lemma IV.1. *Given a BIMS channel $\overline{W}$ with input $\overline{X} \sim P_X$, where $P_X = Ber(1/2)$, and output $\overline{Y}$, we have that*

$$Z_b(\overline{W})^2 \leq \mathsf{H}(\overline{X} \mid \overline{Y}) \leq \log(1 + Z_b(\overline{W})).$$

We thus obtain the following simple corollary.

Corollary IV.1. *Given a BIMS channel W with input $X \sim P_X$, where $P_X = Ber(1/2)$, we have that for the d-view DMC $W^{(d)}$,*

$$Z_b(W)^{2d} \leq \mathsf{H}(X \mid Y^d) \leq Z_b(W)^d.$$

Proof. We first note that when W is a BIMS channel, then so is the d-view DMC $W^{(d)}$. Now, the Bhattacharya parameter

$$\begin{aligned} Z_b(W^{(d)}) &= \sum_{y^d \in \mathcal{Y}^d} \sqrt{P_{Y|0}(y^d|0) P_{Y|1}(y^d|1)} \\ &= \sum_{y^d \in \mathcal{Y}^d} \prod_{i=1}^{d} \sqrt{P_{Y|0}(y_i|0) P_{Y|1}(y_i|1)} \\ &= \prod_{i=1}^{d} \sum_{y_i \in \mathcal{Y}} \sqrt{P_{Y|0}(y_i|0) P_{Y|1}(y_i|1)} \\ &= Z_b(W)^d. \end{aligned}$$

The proof is then completed by appealing to Lemma IV.1 and noting that $\log(1 + z) \leq z$, for all $z > -1$. □

Moreover, we have the following bound on $\rho(I)$ for general binary-input DMCs and for arbitrary input distributions P_X.

Lemma IV.2. *Given a binary-input DMC W with input $X \sim P_X$, where P_X is arbitrary, we have that*

$$\mathsf{H}(X|Y^d) \geq e^{-d \cdot \mathsf{C}(P_{Y|X=0}, P_{Y|X=1}) + \Theta(\log d)}.$$

Proof. Observe that

$$\mathsf{H}(X|Y^d) = \mathbb{E}\left[\log \frac{1}{P_{X|Y^d}(X \mid Y^d)}\right].$$

Further, recall that in a binary Bayesian hypothesis testing setting (see [15, Ch. 11]) with prior P_X and hypotheses $\{P_{Y|0}, P_{Y|1}\}$, we have that the probability of correct decision is given by

$$\mathbb{E}\left[P(X \mid Y^d)\right] = 1 - e^{-d \cdot \mathsf{C}(P_{Y|X=0}, P_{Y|X=1}) + \Theta(\log d)}.$$

Hence, we see that for large enough d,

$$\begin{aligned} \mathsf{H}(X|Y^d) &= \mathbb{E}\left[\log \frac{1}{P(X \mid Y^d)}\right] \\ &\geq -\log\left(1 - e^{-d \cdot \mathsf{C}(P_{Y|X=0}, P_{Y|X=1}) + \Theta(\log d)}\right) \\ &\geq e^{-d \cdot \mathsf{C}(P_{Y|X=0}, P_{Y|X=1}) + \Theta(\log d)}, \end{aligned} \tag{4}$$

where the first inequality follows from Jensen's inequality and the second from the fact that $\log(1 + z) \leq z$, for $z > -1$. Also, the $\Theta(\log d)$ term in the exponent in (4) is due either to the Laplace method (see [25, Sec. 4.2]) or to the results of [26, Thm. 2.1 and Corollary 1]. □

The above ingredients suffice to afford a proof of Theorem III.1 for the special case of BIMS channels with a uniform input distribution.

Proof of Thm. III.1 for BIMS channels with uniform inputs. For the case when W is a BIMS channel, we have that $\mathsf{C}(P_{Y|0}, P_{Y|1}) = -\log Z_b(W)$ (see the remark following the statement of Theorem III.1). Hence, when the input distribution is uniform, we have from Corollary IV.1, we have that $\mathsf{H}(X|Y^d) \leq e^{-d \cdot \mathsf{C}(P_{Y|0}, P_{Y|1})}$. The proof concludes by appealing then to Lemma IV.2, which shows that $\mathsf{H}(X|Y^d) \geq e^{-d \cdot \mathsf{C}(P_{Y|0}, P_{Y|1}) + \Theta(\log d)}$. Hence, overall, we have that $\mathsf{H}(X|Y^d) = e^{-d \cdot \mathsf{C}(P_{Y|0}, P_{Y|1}) + \Theta(\log d)}$. □

We now proceed to proving that Theorem III.1 for general DMCs W.

B. Convergence of Mutual Information for General DMCs

Given a general DMC W with input alphabet $\mathcal{X}$ (not necessarily binary) and output alphabet $\mathcal{Y}$, consider the d-view DMC $W^{(d)}$. Let the input to $W^{(d)}$ be X, with input distribution being P_X, and let its outputs be $Y_1, \ldots, Y_d \in \mathcal{Y}^d$. The joint distribution of the inputs and outputs is such that $P_{X,Y^d}(x, y^d) = P_X(x) \prod_{i=1}^{d} W(y_i|x)$. Much like the proof of Corollary IV.1, the inequality

$$\mathsf{H}(X|Y^d) \leq Z_g(W)^d,$$

can be shown to hold (see [24, Prop. 4.8]), where $Z_g(W) := \sum_{x \neq x'} \sum_{y \in \mathcal{Y}} \sqrt{P(x) W(y|x) P(x') W(y|x')}$ is a scaled version of

the Bhattacharya parameter for general DMCs. This then results in $\rho(I) \geq -\log Z_g(W)$, thereby showing that for well-behaved DMCs, the rate of convergence of $I^{(d)}$ to $H(X)$ is exponential in d. We shall now take up the question of the exact speed of this convergence.

To this end, we will be next interested in evaluating, up to exponential tightness, the behaviour of

$$\begin{aligned}\mathsf{H}(X|Y^d) &= \mathbb{E}\left[\log \frac{1}{P_{X|Y^d}(X \mid Y^d)}\right] \\ &= \sum_{x\in\mathcal{X}} P_X(x)\mathbb{E}\left[\log \frac{1}{P_{X|Y^d}(x \mid Y^d)}\middle| X = x\right]. \end{aligned} \tag{5}$$

Now, fix an $x \in \mathcal{X}$ and focus on the inner term in (5). Then,

$$\begin{aligned}&\mathbb{E}\left[\log \frac{1}{P_{X|Y^d}(x \mid Y^d)}\middle| X = x\right] \\ &\qquad = \int_0^\infty \mathbb{P}\left[\log \frac{1}{P_{X|Y^d}(x \mid Y^d)} \geq t\middle| X = x\right] \mathrm{d}t. \end{aligned} \tag{6}$$

The inner probability, to wit, $p_x(t) := \mathbb{P}\left[-\log P_{X|Y^d}(x \mid Y^d) \geq t | X = x\right]$ is bounded as follows:

Lemma IV.3. *We have that*

$$\frac{p_x(t)}{(|\mathcal{X}|-1)} \leq \max_{\tilde{x}\neq x} \mathbb{P}\left[\frac{P(x)P(Y^d \mid x)}{P(\tilde{x})P(Y^d \mid \tilde{x})} \leq \frac{e^{-(t-\log(|\mathcal{X}|-1))}}{1-e^{-t}}\middle| X = x\right]$$

and

$$p_x(t) \geq \max_{\tilde{x}\neq x} \mathbb{P}\left[\frac{P(x)P(Y^d \mid x)}{P(\tilde{x})P(Y^d \mid \tilde{x})} \leq \frac{e^{-t}}{1-e^{-t}}\middle| X = x\right].$$

Proof. First, observe that

$$\begin{aligned}&p_x(t) \\ &= \mathbb{P}\left[\frac{P_{XY^d}(x, Y^d)}{P_{Y^d}(Y^d)} \leq e^{-t}\middle| X = x\right] \\ &= \mathbb{P}\left[P(x)P(Y^d \mid x) \leq e^{-t}\sum_{\tilde{x}} P(\tilde{x})P(Y^d \mid \tilde{x})\middle| X = x\right] \\ &= \mathbb{P}\left[\frac{P(x)P(Y^d \mid x)}{\sum_{\tilde{x}\neq x} P(\tilde{x})P(Y^d \mid \tilde{x})} \leq \frac{e^{-t}}{1-e^{-t}}\middle| X = x\right]. \end{aligned}$$

We first prove the upper bound in the lemma. The above probability obeys

$$\begin{aligned}&\mathbb{P}\left[\frac{P(x)P(Y^d \mid x)}{\sum_{\tilde{x}\neq x} P(\tilde{x})P(Y^d \mid \tilde{x})} \leq \frac{e^{-t}}{1-e^{-t}}\middle| X = x\right] \\ &\leq \mathbb{P}\left[\frac{P(x)P(Y^d \mid x)}{(|\mathcal{X}|-1)\max_{\tilde{x}\neq x} P(\tilde{x})P(Y^d \mid \tilde{x})} \leq \frac{e^{-t}}{1-e^{-t}}\middle| X = x\right] \\ &= \mathbb{P}\left[\frac{1}{|\mathcal{X}|-1}\cdot\min_{\tilde{x}\neq x}\frac{P(x)P(Y^d \mid x)}{P(\tilde{x})P(Y^d \mid \tilde{x})} \leq \frac{e^{-t}}{1-e^{-t}}\middle| X = x\right] \\ &\leq \sum_{\tilde{x}\neq x}\mathbb{P}\left[\frac{1}{|\mathcal{X}|-1}\cdot\frac{P(x)P(Y^d \mid x)}{P(\tilde{x})P(Y^d \mid \tilde{x})} \leq \frac{e^{-t}}{1-e^{-t}}\middle| X = x\right] \\ &\leq (|\mathcal{X}|-1)\cdot\max_{\tilde{x}\neq x}\mathbb{P}\left[\frac{1}{|\mathcal{X}|-1}\frac{P(x)P(Y^d \mid x)}{P(\tilde{x})P(Y^d \mid \tilde{x})} \leq \frac{e^{-t}}{1-e^{-t}}\middle| X = x\right], \end{aligned}$$

thereby showing the upper bound. Further, for any $\tilde{x}$, we have that

$$\begin{aligned}p_x(t) &= \mathbb{P}\left[\frac{P(x)P(Y^d \mid x)}{\sum_{\tilde{x}\neq x} P(\tilde{x})P(Y^d \mid \tilde{x})} \leq \frac{e^{-t}}{1-e^{-t}} \mid X = x\right] \\ &\geq \max_{\tilde{x}\neq x}\mathbb{P}\left[\frac{P(x)P(Y^d \mid x)}{P(\tilde{x})P(Y^d \mid \tilde{x})} \leq \frac{e^{-t}}{1-e^{-t}} \mid X = x\right], \end{aligned}$$

thereby proving the lower bound. □

Since we are interested mainly in the behaviour of $\mathsf{H}(X|Y^d)$ upto first order in the exponent d, it suffices for us to understand the behaviour of $p_x(t)$ upto first order in the exponent d with the expectation that we will accrue an additional $\Theta(\log d)$ term in the exponent, in the final expression (see Eq. (6) above and [25, Sec. 4.2] for intuition). To this end, following Lemma IV.3, for fixed $z \in [0, \infty)$ and $c > 0$, we shall focus on the term

$$\Gamma_{x,\tilde{x}}(z) := \mathbb{P}\left[\frac{P(x)P(Y^d \mid x)}{P(\tilde{x})P(Y^d \mid \tilde{x})} \leq \frac{ce^{-z}}{1-e^{-z}}\middle| X = x\right],$$

with the reasoning that the rate of convergence of $\mathsf{H}(X|Y^d)$ will be dictated by the smallest rate of $\Gamma_{x,\tilde{x}}(z)$, as z increases from 0 to ∞.

Before we proceed, we require some more notation. Let us denote the log-likelihood ratio by

$$L^y_{x,\tilde{x}} := \log \frac{P_{Y|X}(y \mid x)}{P_{Y|X}(y \mid \tilde{x})}.$$

For a given $y^d \in \mathcal{Y}^d$ and for $b \in \mathcal{Y}$, let $\hat{Q}_{y^d}(b) := \frac{1}{d}\sum_{i=1}^d \mathbb{1}\{y_i = b\}$ be the empirical frequency of the letter b in any output sequence. Further, we define

$$L_{x,\tilde{x}}(\hat{Q}) := \sum_{b\in\mathcal{Y}} \hat{Q}(b)L^b_{x,\tilde{x}}.$$

Finally, let

$$v_{x,\tilde{x}} \equiv v_{x,\tilde{x}}(z) := \frac{1}{d}\left(\log\left(\frac{ce^{-z}}{1-e^{-z}}\right) - \log\frac{P(x)}{P(\tilde{x})}\right)$$

and let $z(v_{x,\tilde{x}}) := \log\left(1 + \frac{cP(\tilde{x})}{P(x)}\cdot e^{-dv_{x,\tilde{x}}}\right)$ be its inverse function.

Proposition IV.1. *We have that for any fixed $\delta > 0$,*

1) *If $v_{x,\tilde{x}} \geq L_{x,\tilde{x}}(P_{Y|x}) + \delta$, then $\Gamma_{x,\tilde{x}}(z(v_{x,\tilde{x}})) = \Theta(1)$.*
2) *If $v_{x,\tilde{x}} \leq L_{x,\tilde{x}}(P_{Y|x}) - \delta$, then $\Gamma_{x,\tilde{x}}(z(v_{x,\tilde{x}})) = e^{-dE(v_{x,\tilde{x}})+\Theta(\log d)}$, where*

$$E(v_{x,\tilde{x}}) := \begin{cases}\max_{\lambda\geq 0}\left[-\log\left(\sum_y P(y|x)^{1-\lambda}P(y|x')^\lambda\right) - \lambda v_{x,\tilde{x}}\right], \\ \qquad\qquad \text{if } v_{x,\tilde{x}} \geq \min_{b\in\mathcal{Y}} L^b_{x,\tilde{x}}, \\ \infty, \text{ otherwise.}\end{cases}$$

Proof. Observe that for any $z \in [0, \infty)$,

$$\begin{aligned}
&\Gamma_{x,\tilde{x}}(z) \\
&= \mathbb{P}\left[\frac{P(x)P(Y^d \mid x)}{P(\tilde{x})P(Y^d \mid \tilde{x})} \le \frac{ce^{-z}}{1-e^{-z}}\middle| X = x\right] \\
&= \mathbb{P}\left[\sum_{i=1}^{d} \log \frac{P_{Y|X}(Y_i \mid x)}{P_{Y|X}(Y_i \mid \tilde{x})} \le \log\left(\frac{ce^{-z}}{1-e^{-z}}\right) - \log\frac{P(x)}{P(\tilde{x})}\middle| X = x\right] \\
&= \mathbb{P}\left[\sum_{i=1}^{d} L_{x,\tilde{x}}^{Y_i} \le \log\left(\frac{ce^{-z}}{1-e^{-z}}\right) - \log\frac{P(x)}{P(\tilde{x})}\middle| X = x\right] \\
&= \mathbb{P}\left[\mathbb{E}_{\hat{Q}}\left[L_{x,\tilde{x}}^{Y^d}\right] \le v_{x,\tilde{x}}(z)\middle| X = x\right].
\end{aligned}$$

Writing this differently, we have that for any $v_{x,\tilde{x}} \in \mathbb{R}$,

$$\Gamma_{x,\tilde{x}}(z(v_{x,\tilde{x}})) = \mathbb{P}\left[\mathbb{E}_{\hat{Q}}\left[L_{x,\tilde{x}}^{Y^d}\right] \le v_{x,\tilde{x}}\middle| X = x\right].$$

Consider first the case when $v_{x,\tilde{x}} \ge L_{x,\tilde{x}}(P_{Y|x}) + \delta$. Note that $L_{x,\tilde{x}}(P_{Y|x}) = D(P_{Y|x}||P_{Y|\tilde{x}}) > 0$. We split this case into two subcases.

- If $v_{x,\tilde{x}} \ge \max_{b\in\mathcal{Y}} L_{x,\tilde{x}}^b$ then $\Gamma_{x,\tilde{x}}(z(v_{x,\tilde{x}})) = 1$.
- Otherwise, consider the case where $L_{x,\tilde{x}}(P_{Y|x}) + \delta \le v_{x,\tilde{x}} \le \max_{b\in\mathcal{Y}} L_{x,\tilde{x}}^b$. Now, observe that

$$\begin{aligned}
1 \ge \Gamma_{x,\tilde{x}}(z(v_{x,\tilde{x}})) \ge \\
\mathbb{P}\left[\mathbb{E}_{\hat{Q}}\left[L_{x,\tilde{x}}^{Y^d}\right] \le L_{x,\tilde{x}}(P_{Y|x}) + \delta\middle| X = x\right].
\end{aligned}$$

  Furthermore, by a simple application of Markov's inequality, we get that since $L_{x,\tilde{x}}(P_{Y|x}) = \mathbb{E}_{Y^d\sim P_{Y^d|x}}[L_{x,\tilde{x}}^{Y^d}]$, the following inequality holds:

$$\mathbb{P}\left[\mathbb{E}_{\hat{Q}}\left[L_{x,\tilde{x}}^{Y^d}\right] \le L_{x,\tilde{x}}(P_{Y|x}) + \delta\middle| X = x\right] \ge \frac{\delta}{L_{x,\tilde{x}}(P_{Y|x}) + \delta}.$$

  Hence, we have that $1 \ge \Gamma_{x,\tilde{x}}(z(v_{x,\tilde{x}})) \ge \frac{\delta}{L_{x,\tilde{x}}(P_{Y|x})+\delta}$.

Hence, overall, if $v_{x,\tilde{x}} \ge L_{x,\tilde{x}}(P_{Y|x}) + \delta$, we get that $\Gamma_{x,\tilde{x}}(z(v_{x,\tilde{x}})) = \Theta(1)$.

Next, consider the case when $v_{x,\tilde{x}} \le L_{x,\tilde{x}}(P_{Y|x}) - \delta$. We again split this case into two subcases.

- If $v_{x,\tilde{x}} \le \min_{b\in\mathcal{Y}} L_{x,\tilde{x}}^b$, then $\Gamma_{x,\tilde{x}}(z(v_{x,\tilde{x}})) = 0$.
- If $\min_{b\in\mathcal{Y}} L_{x,\tilde{x}}^b \le v_{x,\tilde{x}} \le L_{x,\tilde{x}}(P_{Y|X=x})$, then this is a large-deviations event. The probability $\Gamma_{x,\tilde{x}}(z(v_{x,\tilde{x}}))$ thus decays exponentially in d, i.e., $\Gamma_{x,\tilde{x}}(z(v_{x,\tilde{x}})) \doteq e^{-dE_1(v_{x,\tilde{x}})+\Theta(\log d)}$, for some $E_1 : \mathbb{R} \to [0, \infty)$, where by Sanov's theorem (see [15, Sec. 11.4]), we have that $E_1(v_{x,\tilde{x}}) = \min_{Q:\, L_{x,\tilde{x}}(Q)\le v_{x,\tilde{x}}} D(Q \mid\mid P_{Y|X=x})$. We now aim to characterize this function E_1.

  From convex duality (see [27, Sec. 5.2]), we obtain that

$$E_1(v_{x,\tilde{x}}) = \max_{\lambda\ge 0}\min_{Q} D(Q \mid\mid P_{Y|X=x}) + \lambda(L_{x,\tilde{x}}(Q) - v_{x,\tilde{x}}). \tag{7}$$

  First, let us solve the inner minimization. Fix $\lambda \ge 0$. Observe that the term $f(\lambda, Q) := D(Q \mid\mid P_{Y|X=x}) + \lambda L_{x,\tilde{x}}(Q)$ obeys

$$\begin{aligned}
f(\lambda, Q) &= \sum_y Q(y)\left(\log\frac{Q(y)}{P_{Y|X=x}(y)} + \lambda L_{x,\tilde{x}}(y)\right) \\
&= -\sum_y Q(y)\left(\log\frac{P_{Y|X=x}(y)}{Q(y)e^{\lambda L_{x,\tilde{x}}(y)}}\right) \\
&\ge -\log\sum_y \frac{P_{Y|X=x}(y)}{e^{\lambda L_{x,\tilde{x}}(y)}},
\end{aligned}$$

  using Jensen's inequality. Also note that equality holds when $Q(y) = Q^*(y) = \frac{1}{Z_{x,\tilde{x}}(\lambda)} P_{Y|X=x}(y)e^{-\lambda L_{x,\tilde{x}}(y)}$, where $Z_{x,\tilde{x}}(\lambda) := \sum_{y\in\mathcal{Y}} P_{Y|X=x}(y)e^{-\lambda L_{x,\tilde{x}}(y)}$ is the partition function (normalization constant). Substituting Q^* back into the function results

$$\begin{aligned}
f(\lambda, Q^*) &= -\log Z_{x,\tilde{x}}(\lambda) - \lambda v_{x,\tilde{x}} \\
&= -\log\left(\sum_y P_{Y|X=x}(y)e^{-\lambda L_{x,\tilde{x}}(y)}\right) - \lambda v_{x,\tilde{x}} \\
&= -\log\left(\sum_y P_{Y|X=x}^{1-\lambda}(y \mid x)P_{Y|X}^{\lambda}(y \mid \tilde{x})\right) - \lambda v_{x,\tilde{x}}.
\end{aligned}$$

  Plugging back into (7), we get that

$$\begin{aligned}
&E_1(v_{x,\tilde{x}}) \\
&= \max_{\lambda\ge 0}\left[-\log\left(\sum_y P_{Y|X=x}^{1-\lambda}(y \mid x)P_{Y|X}^{\lambda}(y \mid \tilde{x})\right) - \lambda v_{x,\tilde{x}}\right].
\end{aligned}$$

Note that by setting $E(v_{x,\tilde{x}}) = E_1(v_{x,\tilde{x}})$ when $\min_{b\in\mathcal{Y}} L_{x,\tilde{x}}^b \le v_{x,\tilde{x}} \le L_{x,\tilde{x}}(P_{Y|X=x})$ and $E(v_{x,\tilde{x}}) = \infty$ when $v_{x,\tilde{x}} \le \min_{b\in\mathcal{Y}} L_{x,\tilde{x}}^b$, we obtain a tight characterization of the rate of exponential decay of $\Gamma_{x,\tilde{x}}(z(v_{x,\tilde{x}}))$, in these ranges of $v_{x,\tilde{x}}$. □

We now state a prove a lemma that will allow us to identify the exact rate of exponential decay of $\mathsf{H}(X|Y^d)$.

Lemma IV.4. *We have that*

$$\int_0^\infty \max_{\tilde{x}\ne x}\Gamma_{x,\tilde{x}}(z)\mathrm{d}z = e^{-d\cdot\min_{\tilde{x}\ne x}\mathsf{C}(P_{Y|x},P_{Y|\tilde{x}})+\Theta(\log d)}.$$

Proof. Consider an integral of the form

$$\bar{I} = \int_0^\infty \max_{\tilde{x}\ne x}\Gamma_{x,\tilde{x}}(z)\mathrm{d}z.$$

First, observe that

$$\begin{aligned}
\bar{I} &\le \int_0^\infty \sum_{\tilde{x}\ne x}\Gamma_{x,\tilde{x}}(z)\mathrm{d}z \\
&\le |\mathcal{X}|^2\cdot\max_{\tilde{x}\ne x}\int_0^\infty \Gamma_{x,\tilde{x}}(z)\mathrm{d}z \\
&\doteq \max_{\tilde{x}\ne x}\int_0^\infty \Gamma_{x,\tilde{x}}(z)\mathrm{d}z.
\end{aligned}$$

Further,

$$\bar{I} \ge \max_{\tilde{x}\ne x}\int_0^\infty \Gamma_{x,\tilde{x}}(z)\mathrm{d}z.$$

Hence, overall, we have that $\overline{I} = e^{\Theta(\log d)} \cdot \max_{\tilde{x} \neq x} \int_0^\infty \Gamma_{x,\tilde{x}}(z)\mathrm{d}z$. We next focus on the integral

$$I = \int_0^\infty \Gamma_{x,\tilde{x}}(z)\mathrm{d}z,$$

for a fixed $x, \tilde{x}$.

By the definition $v_{x,\tilde{x}} = v_{x,\tilde{x}}(z) = \frac{1}{d}\left(\log\left(\frac{ce^{-z}}{1-e^{-z}}\right) - \log\frac{P(x)}{P(x')}\right)$ and from Prop. IV.1, we see that the integral I can be split into two integrals for its evaluation: integral I_1, when $z \leq \log\left(1 + \frac{cP(\tilde{x})}{P(x)} \cdot e^{-d(L_{x,\tilde{x}}(P_{Y|x})+\delta)}\right) := u(d)$ (or equivalently, $v_{x,\tilde{x}} \geq L_{x,\tilde{x}}(P_{Y|x}) + \delta$), and integral I_2, when $z \geq \log\left(1 + \frac{cP(\tilde{x})}{P(x)} \cdot e^{-d(L_{x,\tilde{x}}(P_{Y|x})-\delta)}\right) := \ell(d)$ (or equivalently, $v_{x,\tilde{x}} \leq L_{x,\tilde{x}}(P_{Y|x}) - \delta$), with $I = I_1 + I_2$. Here, we pick $\delta < L_{x,\tilde{x}}(P_{Y|x})$.

Consider first the integral I_1. We have that

$$\begin{aligned} I_1 &= \int_{z \leq u(d)} \Gamma_{x,\tilde{x}}(z)\mathrm{d}z \\ &= \Theta(1) \cdot u(d). \end{aligned}$$

Now, we claim that $u(d) = e^{-d(L_{x,\tilde{x}}(P_{Y|x})+\delta)+\Theta(\log d)}$. To see this, let $\tilde{c} := \frac{cP(\tilde{x})}{P(x)}$ and observe that from standard identities involving the log function,

$$u(d) \leq \tilde{c}e^{-d(L_{x,\tilde{x}}(P_{Y|x})+\delta)},$$

and

$$u(d) \geq \frac{\tilde{c}e^{-d(L_{x,\tilde{x}}(P_{Y|x})+\delta)}}{1 + \tilde{c}e^{-d(L_{x,\tilde{x}}(P_{Y|x})+\delta)}} \geq \frac{\tilde{c}}{2} \cdot e^{-d(L_{x,\tilde{x}}(P_{Y|x})+\delta)},$$

since $L_{x,\tilde{x}}(P_{Y|x}) = D(P_{Y|x}||P_{Y|\tilde{x}}) > 0$. Hence, overall, we have that for any fixed $\delta > 0$, $I_1 = e^{-d(L_{x,\tilde{x}}(P_{Y|x})+\delta)+\Theta(1)}$. Taking the limit as $\delta \downarrow 0$, we get that

$$I_1 = e^{-dL_{x,\tilde{x}}(P_{Y|x})+\Theta(1)} = e^{-d \cdot D(P_{Y|x}||P_{Y|\tilde{x}})+\Theta(1)}. \tag{8}$$

Consider now the integral I_2. Note that from case 2) in Prop. IV.1, when $z \geq \log\left(1 + \tilde{c}e^{-d \cdot \min_b L^b_{x,\tilde{x}}}\right) := U(d)$, we have that $I_2 = 0$. Hence, we can write

$$I_2 = \int_{\ell(d) \leq z \leq U(d)} \Gamma_{x,\tilde{x}}(z)\mathrm{d}z,$$

where, for this interval $\ell(d) \leq z \leq U(d)$ of interest,

$$\Gamma_{x,\tilde{x}}(z) = e^{-dE(v_{x,\tilde{x}}(z))+\Theta(\log d)},$$

with

$$E(v_{x,\tilde{x}}(z)) = \max_{\lambda \geq 0}\left[-\log\left(\sum_y P(y|x)^{1-\lambda}P(y|x')^\lambda\right) - \lambda v_{x,\tilde{x}}\right], \tag{9}$$

from Prop. IV.1.

It is clear that $E(v_{x,\tilde{x}}(z))$ increases as $v_{x,\tilde{x}}$ decreases. In the integral I_2, we thus make the change of variables

$$v = \frac{1}{d}\left(\log\left(\frac{ce^{-z}}{1-e^{-z}}\right) - \log\frac{P(x)}{P(x')}\right),$$

with $\mathrm{d}z = \frac{-dce^{-dv}}{1+ce^{-dv}}\mathrm{d}v$. Thus,

$$\begin{aligned} I_2 &= dc\int_{v=\min_b L^b_{x,\tilde{x}}}^{L_{x,\tilde{x}}(P_{Y|x})-\delta} e^{-dE(v)} \cdot \frac{e^{-dv}}{1+ce^{-dv}}\mathrm{d}v \\ &= dc\int_{v=0}^{L_{x,\tilde{x}}(P_{Y|x})-\delta} e^{-dE(v)} \cdot \frac{e^{-dv}}{1+ce^{-dv}}\mathrm{d}v + \\ &\qquad dc\int_{v=\min_b L^b_{x,\tilde{x}}}^{0} e^{-dE(v)} \cdot \frac{e^{-dv}}{1+ce^{-dv}}\mathrm{d}v. \end{aligned}$$

Let us call the first integral in the expansion above as $I_{2,a}$ and the second as $I_{2,b}$ (note that $\min_b L^b_{x,\tilde{x}} \leq 0$). Now, observe that

$$\begin{aligned} I_{2,a} &= dc\int_{v=0}^{L_{x,\tilde{x}}(P_{Y|x})-\delta} e^{-dE(v)} \cdot e^{-dv+\Theta(\log d)}dv \\ &= \exp(-d(E(L_{x,\tilde{x}}(P_{Y|x})) + L_{x,\tilde{x}}(P_{Y|x}) - \delta) + \Theta(\log d)). \end{aligned}$$

Here, in the first eqauility, we argue as previously that

$$\frac{e^{-dv}}{1+ce^{-dv}} = e^{-dv+\Theta(1)}.$$

Furthermore, the second equation follows from Laplace's method (e.g., see [25, Sec. 4.2]), using the fact that both $e^{-dE(v)}$ and e^{-dv} increase as v increases. Taking $\delta \downarrow 0$, we obtain that

$$I_{2,a} = e^{-d(E(L_{x,\tilde{x}}(P_{Y|x}))+L_{x,\tilde{x}}(P_{Y|x}))+\Theta(\log d)}. \tag{10}$$

Moreover,

$$\begin{aligned} I_{2,b} &\doteq dc\int_{v=\min_b L^b_{x,\tilde{x}}}^{0} e^{-dE(v)}\mathrm{d}v \\ &\doteq e^{-dE(0)+\Theta(\log d)} = e^{-d \cdot \mathsf{C}(P_{Y|x},P_{Y|\tilde{x}})+\Theta(\log d)}. \end{aligned} \tag{11}$$

Here, for the first equality, we use the fact that when $v \leq 0$,

$$\frac{1}{2} \leq \frac{e^{-dv}}{1+ce^{-dv}} \leq \frac{1}{c},$$

and hence $\frac{e^{-dv}}{1+ce^{-dv}} = \Theta(1)$. The second equality holds again by Laplace's method, using the fact that $e^{-dE(v)}$ increases with v.

Putting everything together, in order to nail down the exponential behaviour of integral I, we need to reconcile Equations (8), (10), and (11), to understand which among I_1, $I_{2,a}$, and $I_{2,b}$, is the largest. First, observe that since $\mathsf{C}(P_{Y|x}, P_{Y|\tilde{x}}) \leq D(P_{Y|x}||P_{Y|\tilde{x}})$, we have that $I_{2,b}$ dominates I_1.

Therefore, the exponential behaviour of integral I is governed by the rate

$$\overline{E} = \min\left\{E(L_{x,\tilde{x}}(P_{Y|x})) + L_{x,\tilde{x}}(P_{Y|x}), \mathsf{C}(P_{Y|x}, P_{Y|\tilde{x}})\right\}.$$

Now note that the maximum over $\lambda \geq 0$ in the expressions for E and C can be rewritten as a maximum over $\lambda \in [0,1]$ (owing to the concavity of $-\log\left(\sum_y P(y|x)^{1-\lambda}P(y|x')^\lambda\right)$ and the fact that it equals 0 at $\lambda = 0, 1$). We then get that

$$E(L_{x,\tilde{x}}(P_{Y|x})) + L_{x,\tilde{x}}(P_{Y|x}) \geq \mathsf{C}(P_{Y|x}, P_{Y|\tilde{x}}),$$

implying that $\overline{E} = \mathsf{C}(P_{Y|x}, P_{Y|\tilde{x}})$. Therefore, $I = e^{-d \cdot \mathsf{C}(P_{Y|x},P_{Y|\tilde{x}})+\Theta(\log d)}$, and hence that $\overline{I} \doteq \max_{\tilde{x} \neq x} e^{-d \cdot \mathsf{C}(P_{Y|x},P_{Y|\tilde{x}})+\Theta(\log d)}$. □

We are now ready to prove Theorem III.1 for the convergence of the mutual information.

Proof of Thm. III.1 for mutual information. Recall from Lemma IV.3 that

$$p_x(t) \le (|\mathcal{X}|-1)\cdot \max_{\tilde{x}\neq x} \Gamma_{x,\tilde{x}}(t),$$

with the constant c in the definition of $\Gamma_{x,\tilde{x}}(t)$ taken to be $|\mathcal{X}|-1$. Thus, from Lemma IV.4 and (6), we get that

$$\mathbb{E}\left[\log \frac{1}{P_{X|Y^d}(x \mid Y^d)}\middle| X=x\right] = \int_0^\infty p_x(t)\mathrm{d}t \le e^{-d\min_{\tilde{x}\neq x} \mathsf{C}(P_{Y|x},P_{Y|\tilde{x}})+\Theta(\log d)},$$

where the inequality holds by the Laplace method [25, Sec. 4.2]. Similarly, since

$$p_x(t) \ge \max_{\tilde{x}\neq x} \Gamma_{x,\tilde{x}}(t),$$

with the constant c in the definition of $\Gamma_{x,\tilde{x}}(t)$ taken to be 1, we get that

$$\mathbb{E}\left[\log \frac{1}{P_{X|Y^d}(x \mid Y^d)}\middle| X=x\right] \ge e^{-d\min_{\tilde{x}\neq x} \mathsf{C}(P_{Y|x},P_{Y|\tilde{x}})+\Theta(\log d)}.$$

Thus,

$$\mathbb{E}\left[\log \frac{1}{P_{X|Y^d}(x \mid Y^d)}\middle| X=x\right] \doteq \max_{\tilde{x}\neq x} e^{-d\cdot\mathsf{C}(P_{Y|x},P_{Y|\tilde{x}})+\Theta(\log d)}.$$

Plugging back in (5), we obtain that $\mathsf{H}(X|Y^d) \doteq \max_{x,\tilde{x}:\tilde{x}\neq x} e^{-d\cdot\mathsf{C}(P_{Y|x},P_{Y|\tilde{x}})+\Theta(\log d)}$, yielding the required result. □

V. Convergence of Channel Dispersion

We now shift our attention to proving the convergence of the dispersion $V^{(d)} = \mathbb{E}\left[\left(\iota(X;Y^d) - I(X;Y^d)\right)^2\right]$, to the varentropy $\mathsf{V}(X)$ of the input distribution P_X, for general DMCs W.

First, we write

$$\begin{aligned}
&V^{(d)}\\
&= \mathbb{E}\left[\left(\log\frac{P(X|Y^d)}{P(X)} - \left(\mathsf{H}(X)-\mathsf{H}(X|Y^d)\right)\right)^2\right]\\
&= \mathbb{E}\left[\left(\log\frac{1}{P(X)} - H(X)\right)^2\right] + \mathbb{E}\left[\left(\log P(X|Y^d) + \mathsf{H}(X|Y^d)\right)^2\right]\\
&\quad + 2\cdot\mathbb{E}\left[(\log P(X|Y^d)+\mathsf{H}(X|Y^d))\cdot\left(\log\frac{1}{P(X)} - H(X)\right)\right]\\
&= \mathsf{V}(X) + \left(\mathbb{E}\left[\left(\log P(X|Y^d)\right)^2\right] - \mathsf{H}(X|Y^d)^2\right) + \theta_d, \qquad (12)
\end{aligned}$$

where we use the fact that $\mathbb{E}\left[-\log P(X|Y^d)\right] = \mathsf{H}(X|Y^d)$, and the cross-term θ_d is

$$2\cdot\mathbb{E}\left[(\log P(X|Y^d)+\mathsf{H}(X|Y^d))\cdot\left(\log\frac{1}{P(X)} - H(X)\right)\right].$$

Further, observe that when $P_X = \mathrm{Unif}(\mathcal{X})$, the cross term vanishes, since for any $x\in\mathcal{X}$, we have that $\log\frac{1}{P(x)} = \log|\mathcal{X}| = \mathsf{H}(X)$. Our proof strategy hence is to first characterize (upto exponential tightness) the behaviour of $\mathbb{E}\left[\left(\log P(X|Y^d)\right)^2\right]$ and then handle the cross-term θ_d.

Lemma V.1. *We have that*

$$\mathbb{E}\left[\left(\log P(X|Y^d)\right)^2\right] \doteq \max_{x,\tilde{x}:\tilde{x}\neq x} e^{-d\cdot\mathsf{C}(P_{Y|x},P_{Y|\tilde{x}})+\Theta(\log d)}.$$

Proof. This proof is entirely analogous to the proof of convergence of $I^{(d)}$ to $\mathsf{H}(X)$, for general DMCs, in Section IV-B, and we only highlight the key steps.

We first consider the term $\mathbb{E}\left[\left(\log P(X|Y^d)\right)^2\right]$. As in (5), we first expand this term as:

$$\mathbb{E}\left[\left(\log P(X|Y^d)\right)^2\right] = \sum_{x\in\mathcal{X}} P(x)\mathbb{E}\left[\left(\log P(X|Y^d)\right)^2\middle| X=x\right],$$

where for any $x\in\mathcal{X}$, we have

$$\begin{aligned}
&\mathbb{E}\left[\left(\log P(x|Y^d)\right)^2\middle| X=x\right]\\
&= \int_{t=0}^\infty \Pr\left[\left(\log P(x|Y^d)\right)^2 \ge t\middle| X=x\right]\mathrm{d}t\\
&= \int_{t=0}^\infty \mathbb{P}\left[\log P(x|Y^d) \ge \sqrt{t}\middle| X=x\right]\mathrm{d}t +\\
&\qquad \int_{t=0}^\infty \mathbb{P}\left[\log P(x|Y^d) \le -\sqrt{t}\middle| X=x\right]\mathrm{d}t\\
&= \int_{t=0}^\infty \mathbb{P}\left[\log P(x|Y^d) \le -\sqrt{t}\middle| X=x\right]\mathrm{d}t.
\end{aligned}$$

Here, the last equality holds since $P(x|Y^d) \le 1$, for any $Y^d\in\mathcal{Y}^d$. As earlier, our task reduces to understanding the exponential behaviour of $q_x(t) := \Pr\left[\log P(x|Y^d) \le -\sqrt{t}\middle| X=x\right]$.

Exactly as in Lemma IV.3, we have that

$$\begin{aligned}
&\frac{q_x(t)}{(|\mathcal{X}|-1)}\\
&\le \max_{\tilde{x}\neq x}\mathbb{P}\left[\frac{P(x)P(Y^d\mid x)}{P(\tilde{x})P(Y^d\mid\tilde{x})} \le \frac{e^{-(\sqrt{t}-\log(|\mathcal{X}|-1))}}{1-e^{-\sqrt{t}}}\middle| X=x\right],
\end{aligned}$$

and

$$q_x(t) \ge \max_{\tilde{x}\neq x}\mathbb{P}\left[\frac{P(x)P(Y^d\mid x)}{P(\tilde{x})P(Y^d\mid\tilde{x})} \le \frac{e^{-\sqrt{t}}}{1-e^{-\sqrt{t}}}\middle| X=x\right].$$

Consider, for a fixed $c>0$ and $z\in[0,\infty)$, the expression

$$\Delta_{x,\tilde{x}}(z) := \mathbb{P}\left[\frac{P(x)P(Y^d\mid x)}{P(\tilde{x})P(Y^d\mid\tilde{x})} \le \frac{ce^{-\sqrt{z}}}{1-e^{-\sqrt{z}}}\middle| X=x\right].$$

As argued in the case of the mutual information, the rate of exponential decay of $\Delta_{x,\tilde{x}}(z)$, as a function of z, dictates the rate of exponential decay of $q_x(t)$, and hence of $\mathbb{E}\left[\left(\log P(X|Y^d)\right)^2\right]$. Once again, letting $v_{x,\tilde{x}} = v_{x,\tilde{x}}(z) = \frac{1}{d}\left(\log\left(\frac{ce^{-z}}{1-e^{-z}}\right) - \log\frac{P(x)}{P(x')}\right)$ and setting $(z(v_{x,\tilde{x}}))^{1/2} := \log\left(1+\frac{cP(\tilde{x})}{P(x)\cdot}e^{-dv_{x,\tilde{x}}}\right)$, we obtain a result similar to Proposition IV.1, with the only modification being the replacement of $\Gamma_{x,\tilde{x}}$ by $\Delta_{x,\tilde{x}}$.

Next, as in Lemma IV.4, consider the integral

$$\bar{I}^V = \int_0^\infty \max_{\tilde{x}\neq x}\Delta_{x,\tilde{x}}(z)\mathrm{d}z.$$

We claim that $\overline{I}^V = e^{-d\min_{\tilde{x}\neq x}\mathsf{C}(P_{Y|x},P_{Y|\tilde{x}})+\Theta(\log d)}$. To obtain this, we argue as in the proof of Lemma IV.4 that $\overline{I}^V \doteq \max_{\tilde{x}\neq x} I^V$, where

$$I^V = \int_0^\infty \Delta_{x,\tilde{x}}(z)\mathrm{d}z,$$

for a fixed $x, \tilde{x}$.

Now, we split the evaluation of I^V into two integrals: integral I_1^V, when $z \leq \left(\log\left(1+\frac{cP(\tilde{x})}{P(x)}\cdot e^{-d(L_{x,\tilde{x}}(P_{Y|x})+\delta)}\right)\right)^2$, and integral I_2^V, when $z \geq \left(\log\left(1+\frac{cP(\tilde{x})}{P(x)}\cdot e^{-d(L_{x,\tilde{x}}(P_{Y|x})-\delta)}\right)\right)^2$, with $I^V = I_1^V + I_2^V$. By arguments analogous to those in the proof of Lemma IV.4, we obtain that

$$I_1^V = e^{-2d\cdot D(P_{Y|x}||P_{Y|\tilde{x}})+\Theta(\log d)},$$

and

$$I_2^V = e^{-d\cdot\mathsf{C}(P_{Y|x},P_{Y|\tilde{x}})+\Theta(\log d)}.$$

Thus, we have that $I^V = e^{-d\cdot\mathsf{C}(P_{Y|x},P_{Y|\tilde{x}})+\Theta(\log d)}$, and hence that $\overline{I}^V = \max_{\tilde{x}\neq x} e^{-d\cdot\mathsf{C}(P_{Y|x},P_{Y|\tilde{x}})+\Theta(\log d)}$. Putting everything togehter, we get that $\mathbb{E}\left[\left(\log P(X|Y^d)\right)^2\right] = \max_{x,\tilde{x}:\tilde{x}\neq x} e^{-d\cdot\mathsf{C}(P_{Y|x},P_{Y|\tilde{x}})+\Theta(\log d)}$. □

Lemma V.2. *We have that*

$$\theta_d = \max_{x,\tilde{x}:\tilde{x}\neq x} e^{-d\cdot\mathsf{C}(P_{Y|x},P_{Y|\tilde{x}})+\Theta(\log d)}.$$

Proof. Recall that the cross term

$$\theta_d := 2\cdot\mathbb{E}\left[(\log P(X|Y^d)+\mathsf{H}(X|Y^d))\cdot\left(\log\frac{1}{P(X)}-H(X)\right)\right].$$

We then write

$$\theta_d(x) := \left(\log\frac{1}{P(x)}-H(X)\right)\cdot\left(\mathbb{E}\left[\log P(x|Y^d)\middle|X=x\right]+\mathsf{H}(X|Y^d)\right),$$

with $\theta_d = \sum_{x\in\mathcal{X}} P(x)\theta_d(x)$. From the capacity analysis, we know that (see (6))

$$\mathbb{E}\left[\log P(x|Y^d)\middle|X=x\right] = -\max_{x'\neq x} e^{-d\cdot\mathsf{C}(P_{Y|x},P_{Y|x'})+\Theta(\log d)}.$$

From our previous analysis, we know that $\mathsf{H}(X|Y^d) = \max_{x,x':x'\neq x} e^{-d\cdot\mathsf{C}(P_{Y|x},P_{Y|x'})+\Theta(\log d)}$. Hence,

$$\theta_d = e^{\Theta(\log d)}\cdot\theta_d(x) = \max_{x,x':x\neq x'} e^{-d\cdot\mathsf{C}(P_{Y|x},P_{Y|x'})+\Theta(\log d)}.$$

□

With these lemmas in place, the proof of Theorem is easily completed.

Proof of Thm. III.1 for dispersion. From (12) we see that

$$V^{(d)} - \mathsf{V}(X) = \mathbb{E}\left[\left(\log P(X|Y^d)\right)^2\right] - \mathsf{H}(X|Y^d)^2 + \theta_d.$$

Putting together Lemmas V.1 and V.2 and the fact that $\mathsf{H}(X|Y^d) = \max_{x,x':x'\neq x} e^{-d\cdot\mathsf{C}(P_{Y|x},P_{Y|x'})+\Theta(\log d)}$ (from the proof of Thm. III.1 for mutual information), we readily obtain the required result. □

VI. Poisson Approximation Channel

In this section, we take up the special case of the d-view BSC(p), denoted by BSC$^{(d)}(p)$. Recall that the capacity of this channel equals the capacity of the binomial channel $\mathsf{Bin}_d(p)$, which is given in (2). Our objective is to show that the Poisson approximation channel $\mathsf{Poi}_d(p)$ introduced in Section III has capacity that is a very good lower bound on the capacity of BSC$^{(d)}(p)$, via the proof of Theorem III.3. Before we proceed with the proof, we state and prove a simple lemma about the concavity of the mutual information $I^{(d)} = I(X;Y^d)$, for any d-view DMC $W^{(d)}$ and input distribution P_X.

Lemma VI.1. *For any d-view DMC $W^{(d)}$ and input distribution P_X, we have that $I^{(d)}$ is concave over the positive integers, i.e.,*

$$I^{(d)} - I^{(d-1)} \geq I^{(d+1)} - I^{(d)},$$

for all $d \geq 1$.

Proof. We first rewrite what we intend to prove differently: since $I^{(m)} = H(X) + H(Y^m) - H(X,Y^m)$, for any $m \geq 0$, with $I^{(0)} := 0$, we would like to show that

$$\begin{aligned} &H(Y^d) - H(Y^{d-1}) + H(X,Y^{d-1}) - H(X,Y^d) \\ &\qquad \geq H(Y^{d+1}) - H(Y^d) + H(X,Y^d) - H(X,Y^{d+1}). \end{aligned} \tag{13}$$

However, observe that for any $m \geq 0$, $H(X,Y^{m+1}) - H(X,Y^m) = H(Y_{m+1}|X,Y^m) = H(Y_{m+1}|X)$, by the channel law of $W^{(d)}$. Hence, (13) reduces to showing that

$$H(Y^d) - H(Y^{d-1}) \geq H(Y^{d+1}) - H(Y^d). \tag{14}$$

Now, by the submodularity of entropy [22, Thm. 1.6], we have that

$$H(Y^{d-1}, Y_{d+1}) - H(Y^{d-1}) \geq H(Y^{d+1}) - H(Y^d).$$

The above implies (14), since the distributions of (Y^{d-1}, Y_{d+1}) and Y^d are equal, by the stationarity of the DMC $W^{(d)}$. □

For clarity, we first write down the capacity $C(\mathsf{Poi}_d(p))$ of the Poisson approximation channel. As argued in Section III, the capacity is achieved when the inputs $X \sim P_X = \mathrm{Ber}(1/2)$, yielding

$$\begin{aligned} &C(\mathsf{Poi}_d(p)) \\ &= \mathsf{H}(X) - \mathsf{H}(X|R_1,R_2) \\ &= 1 + \sum_{\substack{r_1\geq 0,\\ r_2\geq 0}} \frac{e^{-dp}(dp)^{r_1}}{r_1!}\cdot\frac{e^{-d(1-p)}(d(1-p))^{r_2}}{r_2!}\times \\ &\qquad\qquad \log\frac{p^{r_1}(1-p)^{r_2}}{p^{r_1}(1-p)^{r_2}+p^{r_2}(1-p)^{r_1}}. \end{aligned} \tag{15}$$

We now prove Theorem III.3.

Proof. Consider the mutual information between the input and outputs of the $\mathsf{Poi}_d(p)$ channel when P_X is uniform over the inputs. Since the number of outputs $N = R_1 + R_2$ is known to the decoder, we have that

$$\begin{aligned} C(\mathsf{Poi}_d(p)) &= I(X;R_1,R_2) \\ &= I(X;R_1,R_2 \mid N) \\ &= \sum_{n=0}^{\infty}\mathbb{P}[N=n]\cdot I(X;R_1,R_2 \mid N=n). \end{aligned} \tag{16}$$

Now, observe that since the sum of a $\mathrm{Poi}(dp)$ random variable and a $\mathrm{Poi}(d(1-p))$ random variable is distributed according to $\mathrm{Poi}(d)$, we have by symmetry that $N \sim \mathrm{Poi}(d)$. Further, it is easy to check that the conditional distribution $P_{R_1,R_2|N=n}$ obeys $P_{R_1,R_2|N=n}(r_1, r_2 \mid n) = \binom{n}{r_1} p^{r_1}(1-p)^{n-r_1}$, for all $r_1, r_2 \geq 0$ such that $r_1 + r_2 = n$. Hence, we have that $I(X; R_1, R_2 \mid N = n) = C(\mathsf{Bin}_n(p))$.

Thus, substituting in (16), we get that

$$C(\mathsf{Poi}_d(p)) = \mathbb{E}_{N\sim\mathrm{Poi}(d)}\left[C(\mathsf{Bin}_N(p))\right].$$

Next, using the above pleasing expression, we show that $C(\mathsf{Poi}_d(p))$ is indeed a lower bound on $C(\mathsf{Bin}_d(p))$. From Lemma VI.1 we have that $C(\mathsf{Bin}_n(p))$ is concave over the positive integers. Therefore, applying Jensen's inequality, we get that

$$\begin{aligned} C(\mathsf{Poi}_d(p)) &= \mathbb{E}_{N\sim\mathrm{Poi}(d)}\left[C(\mathsf{Bin}_N(p))\right] \\ &\leq C(\mathsf{Bin}_{\mathbb{E}N}(p)) = C(\mathsf{Bin}_d(p)). \end{aligned}$$

Furthermore, we have that

$$\begin{aligned} &C(\mathsf{Bin}_d(p)) - C(\mathsf{Poi}_d(p)) \\ &= C(\mathsf{Bin}_d(p)) - \mathbb{E}_{N\sim\mathrm{Poi}(d)}\left[C(\mathsf{Bin}_N(p))\right] \\ &\leq C(\mathsf{Bin}_d(p)) - 1 + \mathbb{E}_{N\sim\mathrm{Poi}(d)}[Z(p)^N] \\ &\leq \exp(-d(1-Z(p))) - Z(p)^{2d}, \end{aligned}$$

where the two inequalities make use of Corollary IV.1 and the last inequality uses the well-known expression for the moment-generating function of a Poisson random variable. □

VII. Conclusion and Future Work

In this paper, we considered the behaviour of the information rate and channel dispersion of a multi-view discrete memoryless channel (DMC), with an arbitrary input distribution, when the number of views is very large. In particular, we showed that the information rate and the channel dispersion converge exponentially quickly to the input entropy and varentropy, respectively, with an input distribution-independent rate that equals the smallest Chernoff information between two conditional distributions of the output given different inputs. For the special case of the multi-view binary symmetric channel, we also presented a close approximation to its capacity, via the capacity of the so-called Poisson approximation channel.

One interesting direction for future work would be the extension of the techniques and results in this paper to multi-view channels with memory and synchronization channels, such as insertion and deletion channels (see [28] and [29, Sec. VII]).

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